



JSX and State



Learning Objectives

- Build a component that has an array as state and returns JSX that includes mapped state array elements to JSX
- Explain that conditional rendering is a way to display different data on the page according to a condition
- Build a component that has an array as state and conditionally renders that state



Intro

Let's use the [Todo app](#) example again

OPTIONS

Show completed ☒

ADD ITEM

Add

TODO

☐ Go shopping 🗑

☐ Work out 🗑

COMPLETED

☒ See the doctor 🗑



Mapping

```
const incompleteTodos = todos.filter(todo => !todo.completed)
const completedTodos = todos.filter(todo => todo.completed)

<ul className="todo-list">
  {incompleteTodos.map(function(todo) {
    return <li className="todo">{todo.text}</li>
  })}
</ul>
```

Within our component, we can separate our **complete todos** from our **incomplete ones**, and display them on the page by **mapping each item in the array to a JSX element**.



Mapping – Shortened Version

```
const incompleteTodos = todos.filter(todo => !todo.completed)
const completedTodos = todos.filter(todo => todo.completed)

<ul className="todo-list">
  {incompleteTodos.map(todo => <li className="todo">{todo.text}</li>)}
</ul>
```

We can also shorten the previous code by using an **arrow function**. This is a common practice when using JSX.



Time for some live coding

Let's look at rendering a simple array of strings using map



Time for some practice

Use create-react-app

Build your own random dice roll app. A user should be able to:

- See a title "Dice Roll App"
- See five random numbers from 1-6.
- User can click a button to re-roll the dice

Extension

- Pass data via props to a component that's responsible for rendering a die



Conditional rendering

```
const [showCompletedTodos, setShowCompletedTodos] = useState(true)
```

```
{showCompletedTodos && (  
  <section className="completed-list-section">  
    </section>  
)} }
```

Q: What is happening here?



Conditional rendering

```
const [showCompletedTodos, setShowCompletedTodos] = useState(true)

{showCompletedTodos && (
  <section className="completed-list-section">
    </section>
  )}
```

We are relying again on our **showCompletedTodos** state to alter the **behaviour of our UI**. If the state is **true**, then we render the section, otherwise we don't

This technique is called **conditional rendering**

[Here's an article that shows you in more detail how logic operators work in JS](#)



Time for some live coding

Let's update our example to use conditional rendering



Time for some practice

Update your dice roll app

- When the number is a 6, a 🔥 emoji should be displayed next to the number
- When the number is a 1, a 😬 emoji should be displayed next to the number